Notebook

February 25, 2021

0.0.1 Question 0

Question 0A What is the granularity of the data (i.e. what does each row represent)?

Each row represents a record and data about a specific bike share station.

Question 0B For this assignment, we'll be using this data to study bike usage in Washington D.C. Based on the granularity and the variables present in the data, what might some limitations of using this data be? What are two additional data categories/variables that you can collect to address some of these limitations?

Some of the limitations is determining individual bike usage. There is not much data including the individual, so we could add data such as bike share time, or the id of the person how is registered.

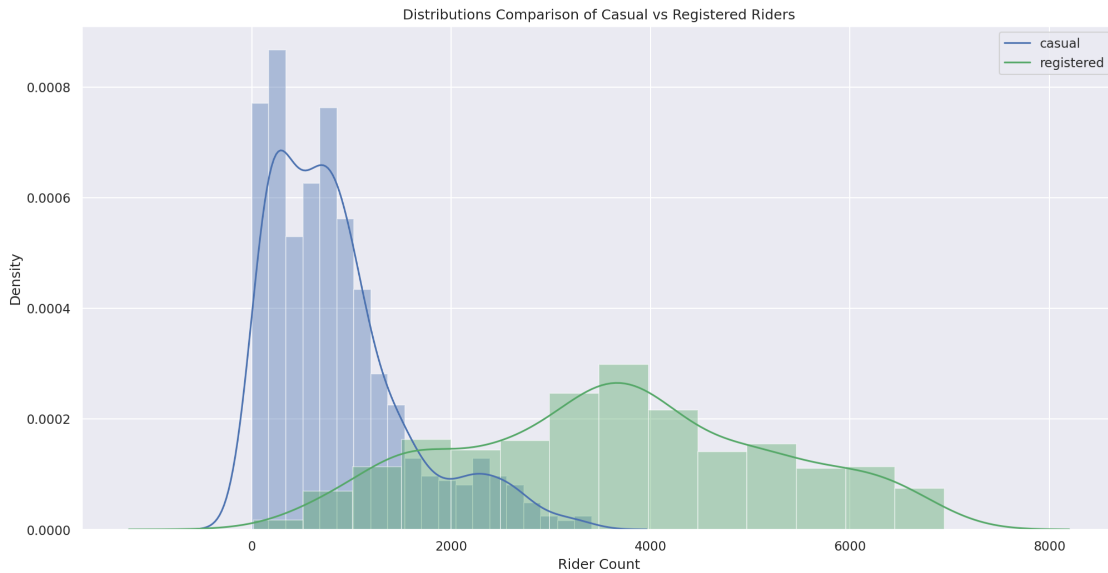
0.0.2 Question 2

Question 2a Use the `sns.distplot` function to create a plot that overlays the distribution of the daily counts of bike users, using blue to represent `casual` riders, and green to represent `registered` riders. The temporal granularity of the records should be daily counts, which you should have after completing question 1c. **You can ignore all warnings that say `distplot` is a deprecated function.**

Include a legend, xlabel, ylabel, and title. Read the seaborn plotting tutorial if you're not sure how to add these. After creating the plot, look at it and make sure you understand what the plot is actually telling us, e.g on a given day, the most likely number of registered riders we expect is ~4000, but it could be anywhere from nearly 0 to 7000.

[116]:
```
plt.title("Distributions Comparison of Casual vs Registered Riders")
sns.distplot(daily_counts['casual'])
sns.distplot(daily_counts['registered'], color='g')
plt.legend(['casual', 'registered'])
plt.xlabel("Rider Count")
```

[116]:
```
Text(0.5, 0, 'Rider Count')
```



0.0.3 Question 2b

In the cell below, descibe the differences you notice between the density curves for casual and registered riders. Consider concepts such as modes, symmetry, skewness, tails, gaps and outliers. Include a comment on the spread of the distributions.

The distribution of casual riders is narrow and skewed to the right, as few people are likely to ride bikes casually. Maybe casual bikers in general own their own bikes and don't need bike shares, explaining the small number in the distribution. But that is an analysis for later. The rider count distribution for the registered riders is much more wider and has higher variance, with an obvious mean at around 4000 riders. The distribution is almost normal, and it is much wider than the causal bikes. This intuitively makes sense as people who frequently use bike share are the ones to register.

0.0.4 Question 2c

The density plots do not show us how the counts for registered and casual riders vary together. Use `sns.lmplot` to make a scatter plot to investigate the relationship between casual and registered counts. This time, let's use the `bike` DataFrame to plot hourly counts instead of daily counts.

The `lmplot` function will also try to draw a linear regression line (just as you saw in Data 8). Color the points in the scatterplot according to whether or not the day is a working day (your colors do not have to match ours exactly, but they should be different based on whether the day is a working day).

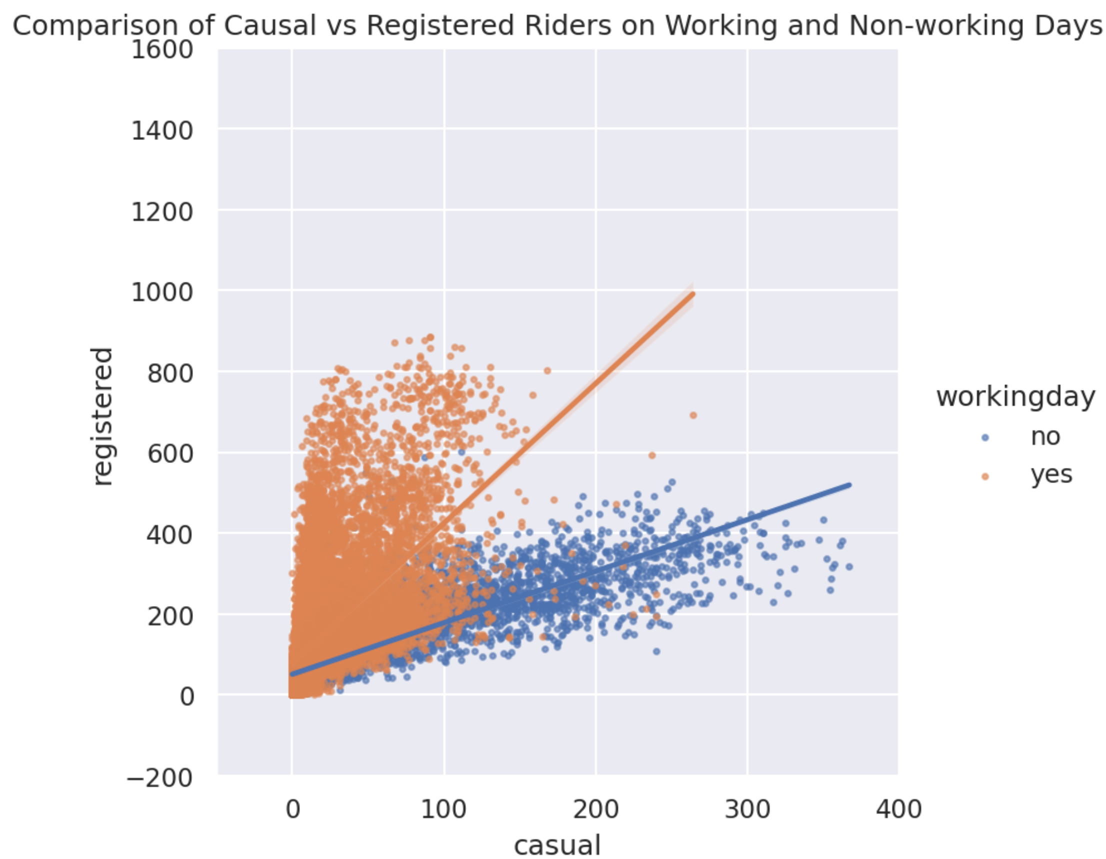
There are many points in the scatter plot, so make them small to help reduce overplotting. Also make sure to set `fit_reg=True` to generate the linear regression line. You can set the `height` parameter if you want to adjust the size of the `lmplot`.

Hints: * Checkout this helpful tutorial on `lmplot`.

- You will need to set `x`, `y`, and `hue` and the `scatter_kws` in the `sns.lmplot` call.
- You will need to call `plot.title` to add a title for the graph.

[117]:
```
# Make the font size a bit bigger
sns.set(font_scale=1)
df = bike[['hr','casual', 'registered', 'workingday']].set_index('hr')
sns_plot = sns.lmplot(x="casual", y="registered", data=df, hue="workingday",␣
↪scatter_kws={"s": 5, "alpha":0.6}, fit_reg=True)
sns_plot.set(xlim=(-50, 400))
sns_plot.set(ylim=(-200, 1600))
plt.title("Comparison of Causal vs Registered Riders on Working and Non-working␣
↪Days")
```

[117]:
```
Text(0.5, 1.0, 'Comparison of Causal vs Registered Riders on Working and Non-
working Days')
```



0.0.5 Question 2d

What does this scatterplot seem to reveal about the relationship (if any) between casual and registered riders and whether or not the day is on the weekend? What effect does overplotting have on your ability to describe this relationship?

The scatterplot reveals that if $y = x$ in the plot, then the ratio between the casual users and registered users is the same. Thus, the relationship inferred from the graph seems to be that casual users are more likely to be bike sharing on non working days, while registered users are more likely to be using bike sharing on working days. It makes sense, as registered users would probably be using bike sharing to commute to work more often. Overplotting on the other hand, makes it hard to see where a lot of users lie or which number best estimates the average of users who are casual or registered. As you can see on the scatter, even setting the alpha to a lower transparency makes it hard to see where most of the data lies.

Generating the plot with weekend and weekday separated can be complicated so we will provide a walkthrough below, feel free to use whatever method you wish if you do not want to follow the walkthrough.

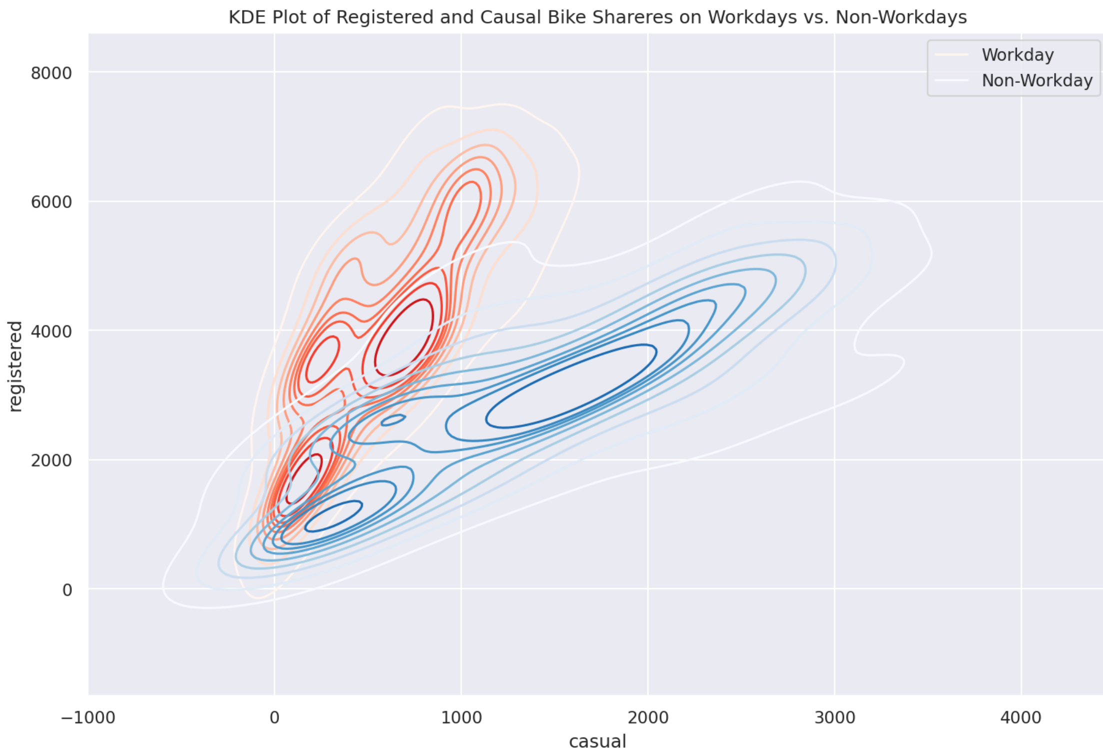
Hints: * You can use `loc` with a boolean array and column names at the same time * You will need to call kdeplot twice. * Check out this guide to see an example of how to create a legend. In particular, look at how the example in the guide makes use of the `label` argument in the call to `plt.plot()` and what the `plt.legend()` call does. This is a good exercise to learn how to use examples to get the look you want. * You will want to set the `cmap` parameter of `kdeplot` to `"Reds"` and `"Blues"` (or whatever two contrasting colors you'd like), and also set the `label` parameter to address which type of day you want to plot. You are required for this question to use two sets of contrasting colors for your plots.

After you get your plot working, experiment by setting `shade=True` in `kdeplot` to see the difference between the shaded and unshaded version. Please submit your work with `shade=False`.

[119]:
```
# Set the figure size for the plot
plt.figure(figsize=(12,8))

# Set 'is_workingday' to a boolean array that is true for all working_days
is_workingday = (daily_counts['workingday'] == 'yes').to_numpy()

# Bivariate KDEs require two data inputs.
# In this case, we will need the daily counts for casual and registered riders␣
↪on workdays
# Hint: consider using the .loc method here.
casual_workday = daily_counts.loc[is_workingday]['casual']
registered_workday = daily_counts.loc[is_workingday]['registered']

# Use sns.kdeplot on the two variables above to plot the bivariate KDE for␣
↪weekday rides
sns.kdeplot(x=casual_workday, y=registered_workday, cmap='Reds')

not_workingday = (daily_counts['workingday'] == 'no').to_numpy()
# Repeat the same steps above but for rows corresponding to non-workingdays
# Hint: Again, consider using the .loc method here.
casual_non_workday = daily_counts.loc[not_workingday]['casual']
registered_non_workday = daily_counts.loc[not_workingday]['registered']
```

```
# Use sns.kdeplot on the two variables above to plot the bivariate KDE for␣
↪non-workingday rides
sns.kdeplot(x=casual_non_workday, y=registered_non_workday, cmap='Blues')
plt.legend(['Workday', 'Non-Workday'])
plt.title("KDE Plot of Registered and Causal Bike Shareres on Workdays vs.␣
↪Non-Workdays")
```

[119]:
```
Text(0.5, 1.0, 'KDE Plot of Registered and Causal Bike Shareres on Workdays vs.
Non-Workdays')
```



Question 3bi In your own words, describe what the lines and the color shades of the lines signify about the data.

The lines can be see as a contour plot, or the density of the plot. The darker the lines, the more denser and more points it has in the specified contour/area. The color is the different classes of data, where the red hues specify a workday, and blue specifies a nonwork day. The legend is not as clear though, as the legend is basing it off the hue.

Question 3bii What additional details can you identify from this contour plot that were difficult to determine from the scatter plot?

I can specifically pinpoint "hot spots" of the data, where most users lie. For example, we see that there are three peaks of interest in the red KDE, while there are two peaks in the blue KDE. These peaks are the densly concentraded areas where bike sharers lies that was not so obvious from the scatterplot, due to overplotting.

0.1 4: Joint Plot

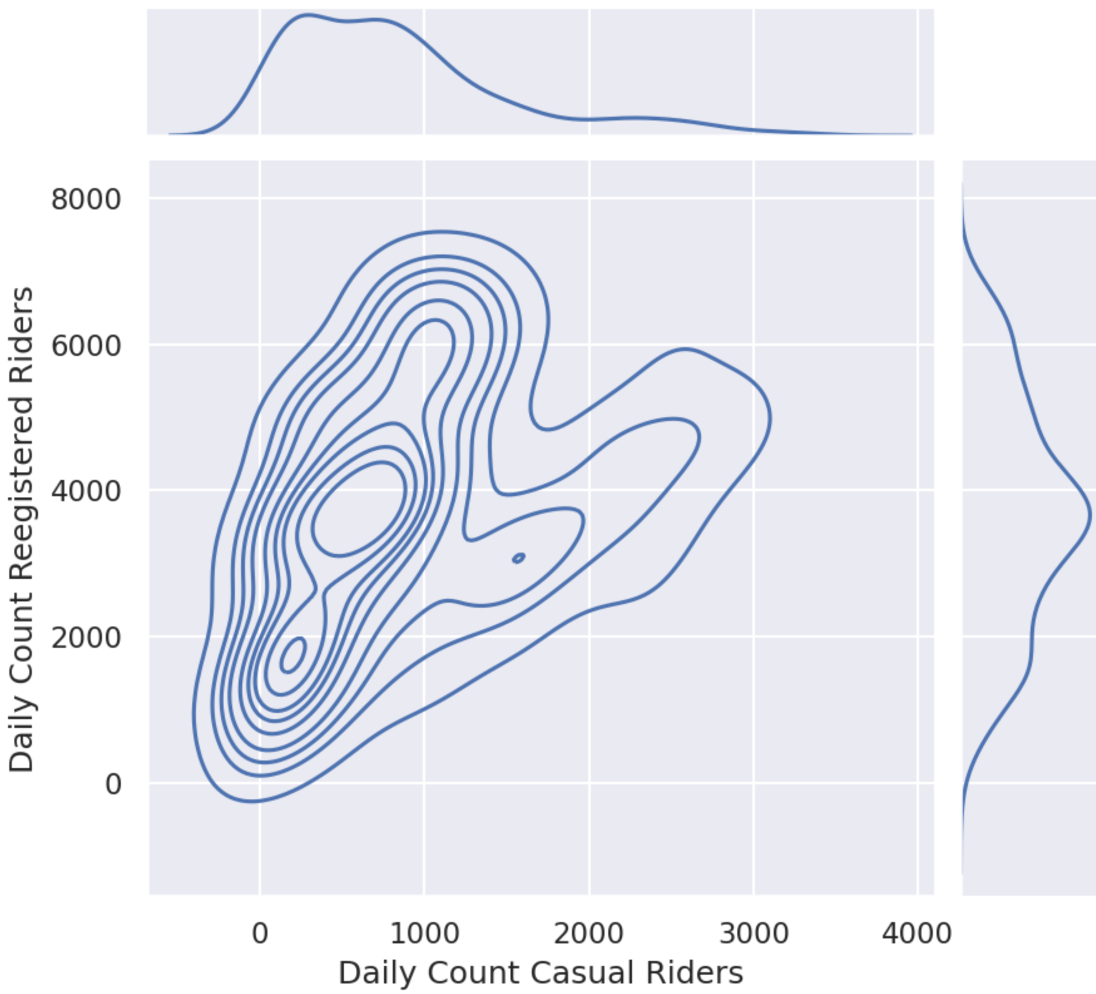
As an alternative approach to visualizing the data, construct the following set of three plots where the main plot shows the contours of the kernel density estimate of daily counts for registered and casual riders plotted together, and the two "margin" plots (at the top and right of the figure) provide the univariate kernel density estimate of each of these variables. Note that this plot makes it harder see the linear relationships between casual and registered for the two different conditions (weekday vs. weekend).

Hints: * The seaborn plotting tutorial has examples that may be helpful. * Take a look at `sns.jointplot` and its `kind` parameter. * `set_axis_labels` can be used to rename axes on the contour plot.

Note: * At the end of the cell, we called `plt.suptitle` to set a custom location for the title. * We also called `plt.subplots_adjust(top=0.9)` in case your title overlaps with your plot.

[120]:
```
fig = sns.jointplot(data=daily_counts, x="casual", y="registered", kind="kde")
plt.suptitle("KDE Contours of Casual vs Registered Rider Count")
fig.set_axis_labels('Daily Count Casual Riders', 'Daily Count Reegistered␣
 ↪Riders')
plt.subplots_adjust(top=0.9);
```

KDE Contours of Casual vs Registered Rider Count



0.2 5: Understanding Daily Patterns

0.2.1 Question 5

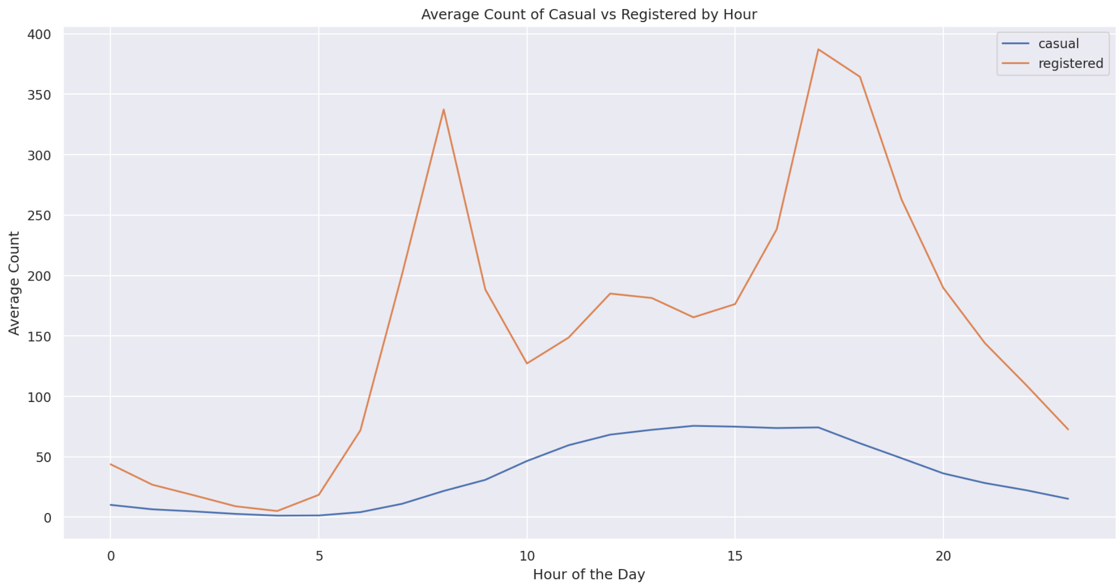
Question 5a Let's examine the behavior of riders by plotting the average number of riders for each hour of the day over the **entire dataset**, stratified by rider type.

Your plot should look like the plot below. While we don't expect your plot's colors to match ours exactly, your plot should have different colored lines for different kinds of riders.

[121]:
```
byhr = bike[['hr', 'casual', 'registered']].groupby('hr').agg('mean')
plt.plot(byhr.index, byhr['casual'])
plt.plot(byhr.index, byhr['registered'])
```

```
plt.legend(['casual', 'registered'])
plt.xlabel('Hour of the Day')
plt.ylabel('Average Count')
plt.title('Average Count of Casual vs Registered by Hour')
```

[121]: Text(0.5, 1.0, 'Average Count of Casual vs Registered by Hour')



Question 5b What can you observe from the plot? Hypothesize about the meaning of the peaks in the registered riders' distribution.

Interestingly, registered users peak at 8 and 17, signifying a 9 to 5 work commute. While casual riders are smoothed out across the afternoon and evening after 10 and 20. Casual riders are probably tourists or bikers enjoying scenic rides in daylight. These are the meanings that I hypothesized about the graph.

In our case with the bike ridership data, we want 7 curves, one for each day of the week. The x-axis will be the temperature and the y-axis will be a smoothed version of the proportion of casual riders.

You should use `statsmodels.nonparametric.smoothers_lowess.lowess` just like the example above. Unlike the example above, plot ONLY the lowess curve. Do not plot the actual data, which would result in overplotting. For this problem, the simplest way is to use a loop.

You do not need to match the colors on our sample plot as long as the colors in your plot make it easy to distinguish which day they represent.

Hints: * Start by just plotting only one day of the week to make sure you can do that first.

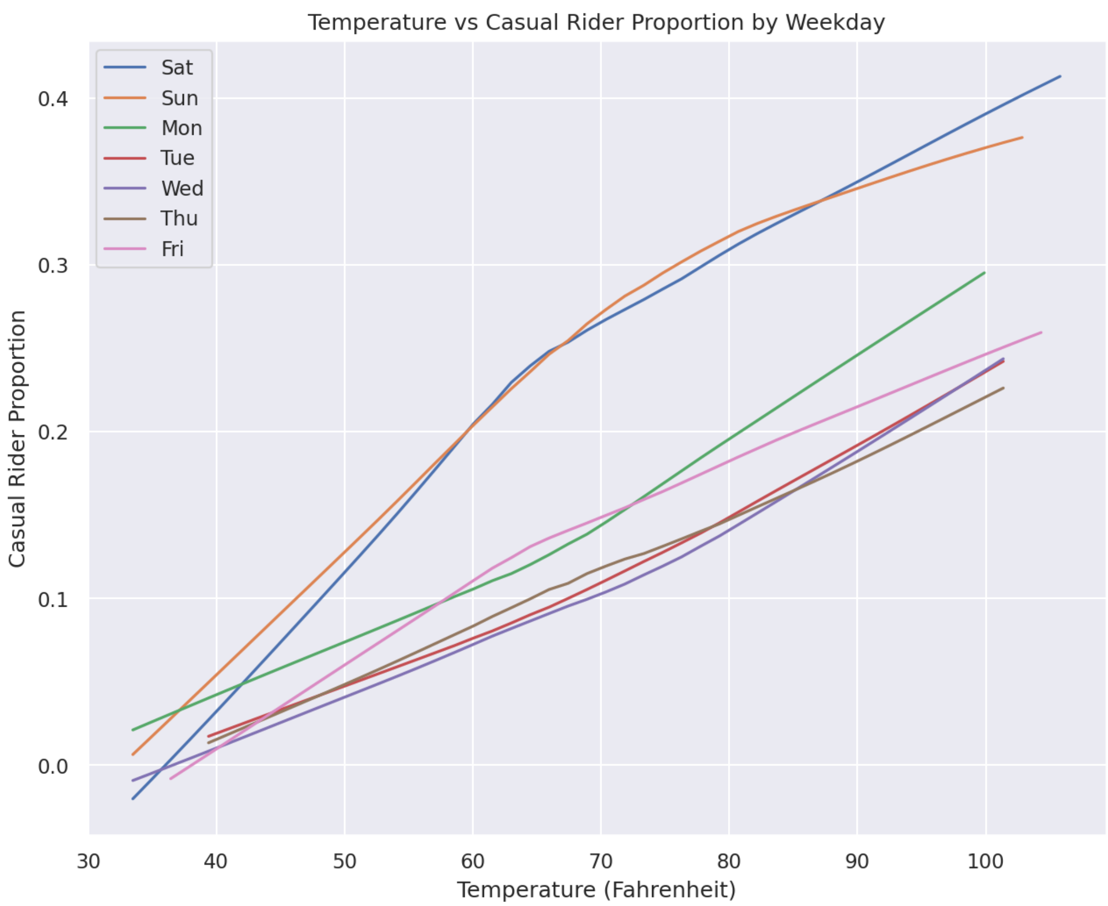
- The `lowess` function expects y coordinate first, then x coordinate. You should also set the `return_sorted` field to `False`.

- Look at the top of this homework notebook for a description of the temperature field to know how to convert to Fahrenheit. By default, the temperature field ranges from 0.0 to 1.0. In case you need it, Fahrenheit = Celsius $* \frac{9}{5} + 32$.

Note: If you prefer plotting temperatures in Celsius, that's fine as well!

[133]:
```python
from statsmodels.nonparametric.smoothers_lowess import lowess

plt.figure(figsize=(10,8))

days = ['Sat', 'Sun', 'Mon', 'Tue', 'Wed', 'Thu', 'Fri']

for day in days:
    bike['real_temp'] = (bike['temp'] * 41.0) * (9.0 / 5.0) + 32
    df = bike.loc[bike['weekday'] == day][['real_temp', 'prop_casual']]
    ysmooth = lowess(df['prop_casual'], df['real_temp'], return_sorted=False)
    sns.lineplot(df['real_temp'], ysmooth, label=day)
plt.title("Temperature vs Casual Rider Proportion by Weekday")
plt.ylabel("Casual Rider Proportion")
plt.xlabel("Temperature (Fahrenheit)")
```

[133]:
```
Text(0.5, 0, 'Temperature (Fahrenheit)')
```



Question 6c What do you see from the curve plot? How is `prop_casual` changing as a function of temperature? Do you notice anything else interesting?

We see that in general, the higher the temperature, the more proportion of casual riders there are. This probably is because casual riders tend to enjoy nicer weather when riding. I also noticed that Saturday and Sunday's lines are much higher than the weekdays. So seems like casual riders like to enjoy riding on the weekend. This could also be an outlier, as riders who have to commute to work are not included, since no one works on weekends, based on our previous observations.

0.2.2 Question 7

Question 7A Imagine you are working for a Bike Sharing Company that collaborates with city planners, transportation agencies, and policy makers in order to implement bike sharing in a city. These stakeholders would like to reduce congestion and lower transportation costs. They also want to ensure the bike sharing program is implemented equitably. In this sense, equity is a social value that is informing the deployment and assessment of your bike sharing technology.

Equity in transportation includes: improving the ability of people of different socio-economic classes, genders, races, and neighborhoods to access and afford the transportation services, and assessing how inclusive transportation systems are over time.

Do you think the `bike` data as it is can help you assess equity? If so, please explain. If not, how would you change the dataset? You may discuss how you would change the granularity, what other kinds of variables you'd introduce to it, or anything else that might help you answer this question.

No, I don't believe that the `bike` dataset helps us access equity. I would add more granlarity to the dataset by adding price of the bike sharing fare, location of sharing inside the city, how far a biker has traveled and returned their bikes from station A to station B, as well as projected income of the area of households, where the bike station lies. These extra variables give us more information about socioeconomic classes and neighborhoods for now, as I feel like genders and races are much more harder data to collect, due to privacy.

Question 7B Bike sharing is growing in popularity and new cities and regions are making efforts to implement bike sharing systems that complement their other transportation offerings. The goals of these efforts are to have bike sharing serve as an alternate form of transportation in order to alleviate congestion, provide geographic connectivity, reduce carbon emissions, and promote inclusion among communities.

Bike sharing systems have spread to many cities across the country. The company you work for asks you to determine the feasibility of expanding bike sharing to additional cities of the U.S.

Based on your plots in this assignment, what would you recommend and why? Please list at least two reasons why, and mention which plot(s) you drew you analysis from.

Note: There isn't a set right or wrong answer for this question, feel free to come up with your own conclusions based on evidence from your plots!

By observing the graph "Distributions Comparison of Casual vs Registered Riders" that I computed, I noticed that there are more registered bike sharers than casual, which may lead to it being less tourism. In order to increase tourism and sightseeing, I think bike sharing is a good way to cause that. So I would advocate for expanding bike sharing to cities that have a lot of tourism or landmarks to make it more enjoyable for travelers/tourists.

After viewing the "After Average Count of Casual vs Registered by Hour" chart, I also noticed that there are more bike sharers on their commute to work. I would also advocate for expansion of bike sharing in busy/metropolin cities. This will hopefully encourage people who live in cities and commute to work across the city to t